MIL-W-3783C
26 August 1986
SUPERSEDING
MIL-W-3783B
29 December 1972

MILITARY SPECIFICATION

WINCH, DRUM, POWER OPERATED, 1-TON PULL, PNEUMATIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a single-drum pneumatic winch.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

RR-W-410	- Wire Rope and Strand.
TT-P-664	- Primer, Coating Synthetic, Rust-Inhibiting, Lacquer-Resisting.
UU-T-81	- Tags, Shipping and Stock.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3950

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:29Z
Check the source to verify that this is the current version before use.

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape: Packaging, Waterproof.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed Flexible.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-F-3541	- Fittings, Lubrication (Hydraulic).
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-838	- Lubrication of Military Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods.

HANDBOOKS

MILITARY

MIL-HDBK-113	- Guide for the Section of Lubricants, Fluids, Preservatives and Speciality Products for Use in Ground Equipment Systems.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:29Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)

AGMA Standards

(Application for copies should be addressed to the American Gear Manufacturers Association, 1330 Massachusetts Avenue, NW, Washington, DC 20005.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME - Boiler Pressure Vessel Code-Welding and Brazing Qualifications Section IX.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1 Structrual Welding Code - Steel.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, NY 10017.)

ANTI-FRICTION BEARING MANUFACTURERS ASSOCIATION (AFBMA)

AFBMA Standard Sections

(Application for copies should be addressed to the Anti-Friction Bearing Manufacturers Association, 60 East 42nd Street, New York, NY 10017.)

INDUSTRIAL FASTENERS INSTITUTE (IFI)

IFI-100 - Prevailing-Torque Type Steel Hexagon Locknuts.
IFI-101 - Torque-Tension Requirements for Prevailing-Torque Type Steel Hexagon Locknuts.

(Application for copies should be addressed to Industrial Fasteners Institute, 1517 Terminal Tower, Cleveland, OH 44113.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE HANDBOOK - Recommended Practices.

J429 - Mechanical and Material Requirements for Externally Threaded Fasteners.
J492 - Rivets and Riveting.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:29Z
Check the source to verify that this is the current version before use.

J614 - Engine and Transmission Dipstick Marking.
J1402 - Automotive Air Brake Hose and Hose Assemblies.
J1500 - Universal Symbols for Operator Controls.

(Application for copies should be addressed to the Society of Automotive Engineers, Two Pennsylvania Plaza, New York, NY 10001.)
(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The winch shall consist of an air-motor-driven, single drum assembled in a complete, compact unit on a mounting base. The winch shall be equipped with a cable and swivel-hook, with safety latch, and shall have a line pull of 2000 pounds in any direction from forward-horizontal to vertical at a speed of not less than 120 feet per minute. The air consumption at the rated load and speed shall be not more than 315 cubic feet per minute (cfm) (60 °F free air) at 80 pounds per square inch (psi) at the inlet. The winch shall be capable of lifting, lowering, and stoping loads within its rated capacity without jerks, jars, or vibrations. The overall dimensions shall not exceed the following:

Length - 40 inches
Width - 24-1/4 inches
Height - 20 inches

The total weight of the assembled unit, less cable, shall be not more than 600 pounds.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of the winch from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish winches conforming to this specification.

3.3 Material. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Threaded parts shall be in accordance with FED-STD-H28. Maximum practicable use shall be made of interchangeable hardware and fastening devices and of a minimum number of types and sizes of studs, bolts, screws, washers, nuts, and similar common (standard) parts. Except as specified herein, flexible hose conforming to SAE J1402, type E or F, with matching reusable fittings, shall be furnished for oil lines.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:29Z
Check the source to verify that this is the current version before use.

3.3.1 Bearings. Bearings used on the winch shall be the antifriction type in accordance with the AFBMA standards. The bearings selected shall conform to AFBMA standards, sections 2 and 3 and coded to section 5. The bearing mountings shall conform to section 7 and the mounting accessories to section 8. The method of bearing load rating shall be in accordance with sections 9 and 10. Any bearings that are not sealed or permanently lubricated will be lubricated through a fitting or by running in a liquid lubricant.

3.3.2 Material deterioration prevention and control. The winch shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the winch may be exposed.

3.3.3 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.4 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.5 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the winch may be newly fabricated from recovered materials to the maximum extent practicable, provided the winch produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the winch.

3.4 Environmental conditions.

3.4.1 Operational temperatures. The winch shall not be damaged during operation in any ambient air temperature between -25 and +125 °F.

3.5 Safety. All exposed parts which are subject to high operating temperatures or which are energized electrically shall be insulated, enclosed, or guarded. All moving parts which are of such nature or so located as to be a hazard to operating or maintenance personnel shall be enclosed or guarded. Protective devices shall not impair the operating functions. All metallic members shall be so assembled and bonded as to provide a continuous electrical path for grounding purposes.

3.6 Performance. The winch shall conform to the performance requirements specified herein without malfunction, damage, or permanent deformation when:

a. The winch mounting base is anchored to an immovable object.
b. The winch is equipped with all attachments, accessories, and equipment when operated as specified herein.
c. The winch is completely serviced and adjusted as specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:29Z
Check the source to verify that this is the current version before use.

3.7 Motor. The motor shall be of the air-driven type with high starting torque, shall be reversible, and shall be complete with combination throttle and reverse valve, lubricator, and governor. It shall be air-enclosed to exclude dust and dirt. The throttle valve shall operate and shall be controlled at the mounted lever. The trottle shall be labled in accordance with SAE J1500 for speed range, if practicle. Otherwise, the throttle control shall be labled "THROTTLE" and "FAST" and "SLOW" at the respective control locations. Normal air pressure for operating the motor shall be 80 psi. The motor shall have a 1-inch pipe connection for inlet air supply.

3.8 Clutch. The clutch shall be designed to slip and hold any cable load up to 2,000 pounds or regulate the speed of the cable drum so that it will reel in or play out cable at any speed between 0 and 36 fpm under the load specified herein. When playing out cable under load with power removed from the winch, the clutch shall function as a brake and shall regulate the drag on the cable drum and the cable speed. This braking function shall also enable the operator to hold any load up to 2,000 pounds without power to the winch. The clutch shall also incorporate a dead-man brake capable of holding any load up to 2,000 pounds with or without power applied to the winch. The dead-man brake shall be so designed that when the clutch-control lever is released by the operator, the brake will be automatically actuated.

3.8.1 Clutch control. A single clutch-control lever shall be conveniently located on the winch in such a manner as to facilitate ease of operation. The force required to operate the clutch-control lever with any load up to 2,000 pounds shall not exceed 25 pounds. The clutch control shall be so designed that when the winch operator releases the lever, the dead-man brake specified in 3.8 will be actuated. The clutch control shall be labled "CLUTCH" and "ON" and "RELEASE" at the respective control locations.

3.9 Brake. The brake shall be of the external contracting (band) type and shall be accessible for inspection and adjustment. The brake bands should be mounted so not more than six bolts are required to remove and replace and no assemblies are required to be removed. The brake shall be operated by a lever located on the winch within reach of the operator. The brake shall stop and hold a 2,000-pound rated load in any position with the drum spooled to capacity with cable. In addition, a positive locking pawl and ratchet arrangement shall be provided for holding loads.

3.10 Winch drum and wire rope. The winch drum shall be of cast or fabricated steel of equal strength and hardness. The brake drum (drums) and winch drum flanges shall be cast or molded integrally with the winch drum. The winch drum, flanges, and braking surfaces shall be accurately machined to provide for intended performance and cable protection. The winch drum shall have a capacity of not less than 625 feet of 3/8-inch wire or 350 feet of 1/2-inch wire rope. A 350-foot length of 1/2 inch, uncoated, regular lay, preformed wire rope, conforming to RR-W-410, type XXIII, shall be furnished wound on the drum. A forged-steel swivel-hook with a minimum clear opening of 1/2 inch shall be furnished fastened to the free end of the wire rope by means of a steel thimble and four wire-rope clamps. The swivel-hook shall be accurately machined at the bearing surface to permit smooth operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:29Z
Check the source to verify that this is the current version before use.

3.11 Gears. All gears and pinions shall be of one-piece steel construction with cut teeth conforming to AGMA Standards. Gears shall be tightly fitted and keyed to their respective shafts.

3.12 Frame or housing. The frame or housing shall be designed to provide the strength and rigidity required in winching operations. The frame or housing shall contain the other components of the winch in one compact unit which shall meet the dimensional and weight requirements specified in 3.1. The frame or housing shall have a flat mounting base in which not less than four holes are drilled, adequately spaced and sized for bolting the winch to a foundation.

3.13 Lubrication. All surfaces requiring lubrication shall be provided with a suitable means for lubricating.

3.13.1 Lubricants. The winch shall operate as specified herein when lubricated with standard military lubricants (see 6.5). The procedure for the selection of lubricants shall be in accordance with MIL-STD-838, section 5. Lubricants selected shall be in accordance with MIL-HDBK-113, chapter 2. When the specification of the lubricant selected includes a requirement for a qualified products list (QPL), the lubricant supplied shall be from a source that is listed on the applicable QPL.

3.13.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, type I, II, or III. Fittings shall be located in a protected position. Labels shall be provided to identify location of fittings which are not readily apparent. Fittings shall be accessible to a grease gun conforming to MIL-G-3859. Accessibility to fittings shall be provided without removing or adjusting accessories or parts. Plates equipped with hand-operable, quick-disconnect fasteners may be removed.

3.13.3 Pressure-release device. Where the use of 1,000 psi and higher pressure-lubricating equipment will damage grease seals or other parts, a suitable pressure-release device shall be provided.

3.13.4 Enclosures with integral reservoir of lubricant. Enclosures such as gear cases which contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, fingerholes, or sightholes to determine the adequacy of the level of the lubricant. Dipstick marking shall be in accordance with SAE J614. Each enclosure shall be equipped with a drainplug and means for filling the enclosure with lubricant. The drainplug shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be into a collecting vessel when the equipment is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain into collecting vessel. Accessibility to the drainplug, the filling means, and the lubricant level checking device shall be provided without removing or adjusting accessories or parts. Plates equipped with hand-operable, quick-disconnect fasteners may be removed.

3.13.5 Initial lubrication service. Initial grease lubrication including lubrication of sealed bearings, shall be with standard military thermally stable greases conforming to MIL-G-10924. The equipment shall be serviced with winter grades of standard military oils designated for use in the temperature range of

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:29Z
Check the source to verify that this is the current version before use.

0 to +32 °F before delivery to the Government (see 6.5). After lubrication service, each item of equipment shall be tagged in a conspicuous place to indicate the temperature range and grades of oils and greases used.

3.14 Slinging provisions. Permanently affixed slinging provisions that enable the winch to be lifted in its normal travel position shall be provided and shall conform to MIL-STD-209. Slinging provisions shall be labled "LIFT HERE" in one inch or greater letter height. Slinging provisions may also be used as tiedowns when such provisions meet the requirements specified in 3.15.

3.15 Tiedown provisions. Permanently affixed provisions that permit fastening the winch to the floor or deck of a transportation medium shall be provided and shall conform to MIL-STD-209. Tiedown provisions shall be labled "TIEDOWN" in one inch or greater letter height. Tiedown provisions may also be used as slinging provisions when such provisions meet the requirements specified in 3.14.

3.16 Identification marking. The winch shall be identified in accordance with MIL-STD-130. The marking shall be applied to the winch on plates conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.17 Instruction plates. Each winch shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the winch. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.18 Treatment and painting. The portions of the winch and its components and parts normally painted in commercial practice shall be cleaned, treated, and painted in accordance with MIL-T-704, type A, color as specified (see 6.2).

3.19 Toolbox. The equipment shall be provided with a toolbox made of metal not less than 0.073 inch in thickness. The toolbox shall have a hinged lid and a latch with a provision for locking the lid that will keep the lid closed when subjected to vibration. The toolbox shall be securely fastened to the equipment in a protected position. A hole shall be provided in the bottom of the toolbox to allow for drainage of excess moisture.

3.20 Resistance to saline atmosphere. When resistance to saline atmosphere is specified (see 6.2), all components shall be designed and installed to withstand the corrosive effect of the salt fog test specified in 4.5.2 without any loss of performance and without sustaining harmful corrosion.

3.21 Workmanship. The winch shall be free from workmanship deficiencies that could impair the operation or serviceability of the winch. All parts, components and assemblies of the winch, including bearings, seals, machinery, stamping, welded parts, castings, forgings, and machined work, shall be clean and free from sand, dirt, fins, pits, sprues, scales, flux, and other harmful, extraneous material. External surfaces shall be free of burrs and sharp edges and corners, except when sharp edges and corners are functional.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:29Z
Check the source to verify that this is the current version before use.

3.21.1 Welders and welding.

3.21.1.1 Welders. Before assigning any welder or welding operator to manual welding work covered by this specification, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed by either of the following codes for the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1 Structural Welding Code - Steel.

ASME - Boiler Pressure Vessel Code-Welding and Brazing Qualifications, Section IX.

The certification shall be made available for review by the contracting officer or the contracting officer's representative. The Government reserves the right to require the recertification of any welder or welding operator.

3.21.1.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are tested as specified herein.

3.21.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have burrs removed. Washers or lockwashers shall be provided. Self-locking nuts are acceptable in lieu of standard nuts and lockwashers. All capscrews, bolts, and nuts shall be tight and correctly torqued. Critical load carrying bolted connections shall, in addition, be as specified herein. Boltholes shall be reamed as required. All critical junctures and joints shall be fastened with high strength bolts with washers and nuts, hex or 12-point type. Bolts shall be not less than SAE J429, grade 5, and shall be of higher grade as necessary. Integral or separate washers shall be of the flat hardened steel type, sized to spread the clamping loads such that brinnelling or embedding in the component members shall be prevented. Mating nuts shall be of the self- locking type in accordance with IFI Standards 100 and 101. Fasteners shall be lubricated prior to wrenching, and installation tools shall be proofed at set intervals to insure constant and correct torque performance.

3.21.3 Riveted connections. Riveted connections shall be in accordance with SAE J492. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.21.4 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified herein. Unless prior approval is obtained from the contracting officer, all holes requiring tapping shall be countersunk before tapping.

3.21.5 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:29Z
Check the source to verify that this is the current version before use.

defined as two or more parts possessing such functional and physical characteristics to be equivalent in performance and durability and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts except for adjustments and without selection for fit or performance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and regulations.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection. Prior to examination and test of the winch, the following operations shall be performed:

a. Winch serviced with oils and greases designated for use in the ambient temperature at which the test will be run.
b. Winch run-in as prescribed by the contractor.

No maintenance, except as prescheduled by the supplier's maintenance publications shall be performed on the winch throughout the tests specified herein.

4.3.1 Examination. The winch shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:29Z
Check the source to verify that this is the current version before use.

4.3.2 Tests. The winch shall be tested as specified in 4.5.2.1 through 4.5.2.4. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each winch shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each winch shall be tested as specified in 4.5.2.1 and 4.5.2.3. Failure of any test shall be cause for rejection.

4.5 Inspection procedures.

4.5.1 Examination. The winch shall be examined for the following defects:

101. Description not as specified.
102. Material not as specified.
103. Bearings not as specified.
104. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.
105. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
106. Contractor does not have documentation available for identification of material, material finishes, or treatment.
107. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the winch.
108. Safety not as specified.
109. Motor not as specified.
110. Clutch not as specified.
111. Clutch control not as specified.
112. Brake not as specified.
113. Winch drum not as specified.
114. Wire rope not as specified.
115. Gears not as specified.
116. Frame or housing not as specified.
117. Lubrication not as specified.
118. Lubrication fittings not as specified.
119. Pressure-release device not as specified.
120. Enclosures with intergral reservoir of lubricant not as specified.
121. Initial lubrication service not as specified.
122. Slinging provisions not as specified.
123. Tiedown provisions not as specified.
124. Identification markings not as specified.
125. Instruction plates not as specified.
126. Treatment and painting not as specified.
127. Toolbox not as specified.
128 Resistance to saline atmosphere not as specified.
126. Workmanship not as specified.
130. Welder certification not as specified.
131. Welding not as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:29Z
Check the source to verify that this is the current version before use.

132. Bolted connections not as specified.
133. Riveted connections not as specified.
134. Machine work not as specified.
135. Interchangeability not as specified.

4.5.2 Tests. The first article model shall be tested as follows. Failure to pass any test shall be cause for rejection.

4.5.2.1 Performance test. The winch shall be bolted down to a suitable foundation and A-frame (or any structure high and strong enough to hold a hook block not less than 35 feet above the test load), and the ability of the winch to lift, lower, and stop a 2,000-pound load shall be determined for both forward-horizontal linepull and vertical linepull (see 3.1). The test load shall be hoisted with one line not less than 30 feet, lowered and stopped 5 to 10 feet from the ground, and then lowered to the ground. This shall constitute one cycle. This test shall be repeated for a total of 1000 cycles. Air supply shall be provided by a 315-cfm or larger compressor. Air pressure at the winch motor and the air consumption shall be not more than that specified in 3.1. Minimum hoisting speed shall be determined during the above test to ascertain if it meets the speed required in 3.1. Nonconformance to 3.1 shall constitute failure of this test.

4.5.2.2 Environmental tests. The unit shall be subjected to a temperature of -25 °F for a period of not less than 4 hours. After this 4-hour period, the unit shall be operated and used in winching operations at -25 °F. This test shall be repeated at a temperature of +125 °F. Nonconformance to 3.4.1 shall constitute failure of this test.

4.5.2.2.1 Salt fog test. When the winch is specified for resistance to corrosion in saline atmosphere, the winch shall be tested in accordance with MIL-STD-810, test method 509, Salt Fog. The winch shall not be washed or operated in accordance with method 509 until the exposure and drying periods have been completed. Nonconformance to 3.20 shall constitute failure of this test.

4.5.2.3 Brake test. The winch brake shall be tested under maximum load (see 3.9). The maximum load shall be lifted to a height of 20 feet and then allowed to drop intermittently at regular speed intervals. Three stops shall be made during each 20-foot drop holding the load until the brake is released manually. This test shall be repeated 10 times in 10 minutes and no overheating shall result from test. Nonconformance to 3.9 shall constitute failure of this test.

4.5.2.4 Tiedown and lifting attachments. The static load as specified in 3.14 for lifting attachments and in 3.15 for tiedown attachments shall be applie to the attachment when mounted on its structural member. Weld failure or permanent deformation of the attachment or the structural member shall constitut failure of this test.

4.6 Inspection of packaging.

4.6.1 First article pack inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:29Z
Check the source to verify that this is the current version before use.

4.6.1.1 Examination. Examine the first article pack for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

136. Keys removed not secured to the keyway of the primary component or shaft for level A.
137. Matchmarking information not applied with a waterproof ink as specified for level A.
138. Unpainted surfaces not coated with preservative as specified for level A.
139. Exposed gears not preserved as specified for level A.
140. Gear housing not filled and tagged as specified for level A.
141. Clutches not preserved and secured in the disengaged position as specified for level A.
142. Wire rope not coated with preservative and wound and secured on the drum as specified for level A.
143. Air motor not preserved as specified for level A.
144. Instructional manual not preserved as specified for level A or C.
145. Repair parts not preserved in accordance with the referenced document as specified for level A or C.
146. Maintenance tools not preserved in accordance with the referenced document as specified for level A or C.
147. Consolidation not as specified for level A.
148. Blocking, bracing and anchoring not in accordance with the referenced document for level A.
149. Strapping not zinc coated for level A.
150. Marking missing, illegible, incorrect or incomplete.

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination within the time frame specified (see 6.2), to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The first article pack may be accomplished utilizing either the first article winch or a production winch. If the first article model winch is utilized, any preservation, and packing shall be removed by the contractor at no expense to the Government when requested by the Government to facilitate comparison between the first article model winch and the production winch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:29Z
Check the source to verify that this is the current version before use.

5.2 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Disassembly. Disassembly shall be the minimum necessary to safeguard parts known to be subject to damage or loss. Bolts, nuts, pins, and washers involved shall be placed on one of the mating parts and secured to prevent their loss. Keys removed shall be secured to the keyway of the primary component or shaft.

5.2.1.2 Matchmarking. Parts that are removed shall be matchmarked when necessary to facilitate reassembly. Matchmarking information shall be on shipping tags conforming to UU-T-81, type A, and shall be applied using a waterproof ink.

5.2.1.3 Preservatives. Preservatives specified herein shall conform to the applications listed in and shall be applied in accordance with MIL-P-116.

5.2.1.4 Unprotected surfaces. Unpainted exterior metal surfaces of components requiring the application of a contact preservative in acordance with MIL-P-116, and not specifically specified herein shall be coated with type P-1 preservative.

5.2.1.5 Gears.

5.2.1.5.1 Exposed gears. Exposed gears shall be coated with type P-1 preservative or with primer conforming to TT-P-664.

5.2.1.5.2 Enclosed gears. Gear housings shall be filled to the operating level with the lubricant recommended for operation and the gears actuated to insure coating all interior parts and surfaces of the housing with the gear lubricant. A tag conforming to UU-T-81, type A, shall be attached in a conspicuous location indicating: "The housing is filled to operating level with lubricant required for operation." The information on the tag shall be applied using a waterproof ink.

5.2.1.6 Clutches.

5.2.1.6.1 Disc-type clutches. With the cover plates removed and the clutch engaged, all accessible metal interior components of the clutch shall be sprayed with a thin film of primer conforming to TT-P-664. Coating of nonmetallic surfaces shall be kept at a minimum. Clutch control mechanisms not enclosed shall be coated with type P-1 preservative. After being sprayed, spring-loaded-type clutches shall have the clutch lever secured in a partially disengaged position to eliminate contact between the disc facings and pressure plates. Snap-over-center clutches and toggle-in-type clutches shall be completely disengaged.

5.2.1.6.2 Cone type. The unlined cone and clutch control mechanisms shall be coated with a thin film of primer conforming to TT-P-664. The clutch shall be disengaged to prevent the clutch lining from adhering to the unlined cones. Clutch control mechanisms not enclosed within housings shall be coated with type P-1 preservative.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:29Z
Check the source to verify that this is the current version before use.

5.2.1.6.3 Jaw type. Clutch jaws, shifter yokes, and machined surfaces of shafts shall be coated with type P-4 preservative. Clutch control mechanisms enclosed within housings shall be fogged with lubricating oil conforming to MIL-L-21260, grade 30. Clutch control mechanisms not enclosed within housings shall be coated with type P-1 preservative. The clutch shall not be operated after application of preservatives.

5.2.1.6.4 Band and shoe type. Clutch drum facings shall be coated with primer conforming to TT-P-664. Clutch control mechanisms enclosed within housings shall be fogged with lubricating oil conforming to MIL-L-21260, grade 30. Clutch control mechanisms not enclosed in the housings shall be coated with type P-1 preservative, or with primer conforming to TT-P-664. Preservative shall not contact the clutch lining.

5.2.1.7 Motor. Interior surfaces of the air motor shall be coated with lubricating oil conforming to MIL-L-21260, grade 30, applied through the lubricating system; or while the motor is operating, the oil shall be fed into the air inlet until the oil appears at the exhaust ports. Air inlets and outlets shall be closed with plastic or metal caps or plugs or shall be covered with barrier material conforming to MIL-B-121, type I or II, grade A, class 1 or 2, secured in place with tape conforming to PPP-T-60, type IV.

5.2.1.8 Winch drum, wire rope, fittings, and swivel hook. The wire rope and unpainted surfaces of the winch drum, fittings, and swivel hook shall be coated with type P-1 preservative. The wire rope shall be evenly wound on the winch drum, and the swivel hook on the end of the wire rope shall be secured to the drum with twisted, soft annealed wire, to prevent unwinding of the rope.

5.2.1.9 Instructional manuals. Instructional manuals shall be preserved in accordance with MIL-P-116, method IC-3.

5.2.1.10 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve the repair parts, when they are included.

5.2.1.11 Maintenance tools. Maintenance tools, when included, shall be preserved in accordance with the level A requirements of PPP-P-40.

5.2.1.12 Consolidation. Preserved small components shall be placed in the toolbox, as space permits, and the toolbox lid closed and secured. Components that will not fit in the toolbox shall be consolidated packaged together in close-fitting boxes conforming to PPP-B-636, W5c or W6c, as applicable to the weight of contents.

5.2.2 Level C.

5.2.2.1 Repair parts, maintenance tools, and instructional manuals. The repair parts, maintenance tools, and instructional manuals shall be preserved as specified for level A and placed in the toolbox. The toolbox lid shall be closed and secured.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:29Z
Check the source to verify that this is the current version before use.

5.2.2.2 Other components. All other components of the winch shall be preserved to afford protection against deterioration and damage from the contractor to the initial destination.

5.3 Packing. Packing shall be level A or C, as specified (see 6.2).

5.3.1 Level A. Each complete winch, preserved as specified in 5.2, shall be packed in a close-fitting wood-cleated, plywood box conforming to PPP-B-601, overseas type, style A, B, or I. Blocking, bracing, and anchoring of the contents shall be in accordance with MIL-STD-1186. Strapping shall be zinc coated.

5.3.2 Level C. Each complete winch, preserved as specified in 5.2, shall be packed in a suitable container which will protect the item from damage and will allow efficient handling during shipment and limited storage.

5.4 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The winch is intended for use in general winching operations.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Time frame required for submission of first article model (see 3.2).
c. When the Government will conduct any or all of the first article model examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
d. Color required (see 3.18).
e. When resistance to saline atmosphere is required (see 3.20).
f. Time frame required for submission of first article pack (see 5.1).
g. Level of preservation and packing required (see 5.2 and 5.3).

6.3 First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of 2 units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.4 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, package, pack, and mark the winches in accordance with this specification.

6.5 Lubricants. MIL-STD-838, Lubrication of Military Equipment, prescribes the policy for using specification-type products wherever possible and provides

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:29Z
Check the source to verify that this is the current version before use.

specific requirements for potential use of non-standard proprietary products. MIL-STD-838 is implemented by MIL-HDBK-113, Guide for the Selection of Lubricants, Fluids, Preservatives and Speciality Products for Use in Ground Equipment Systems. The contracting officer should note that unless otherwise authorized by the US Army Belvoir Research, Development and Engineering Center (ATTN: STRBE-VF), Fort Belvoir, VA 22060-5606, lubricants, fluids, and greases for ground equipment systems must be restricted to those listed under MIL-HDBK-113, chapter 2.

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of winches. A suggested paragraph is as follows:

"Shipment of winches shall include repair parts, maintenance tools, operational instructions, and accessories, unless exceptions are provided elsewhere in the contract."

6.7 Subject term (keyword) listing

One-ton pull.
Pneumatic-motor-driven
Single drum
Swivel-hook
Winch
Wire rope

6.8 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Air Force - 99

Review activity:
Air Force - 84

Preparing activity:
Army - ME

Project 3950-0216

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:29Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

DEPARTMENT OF THE ARMY
U.S. Army Belvoir Research,
Development and Engineering Center
ATTN: STRBE-TSE
Fort Belvoir, VA 22060-5606

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:29Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-3783C	Winch, Drum, Power Operated, 1-Ton Pull, Pneumatic

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:29Z
Check the source to verify that this is the current version before use.